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Roll No

CE-8047 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Advance Water Resources Engineering
(Elective - II)
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Discuss the different parameters considered by a decision maker for water resources allocation in a multiobjective water resources project.
2. a) What is importance of linear programming in system analysis of water resources.
b) Explain Reservoir operation.
3. a) Explain flood routing through reservoirs.
b) Explain stochastic models in hydrology:
 - i) AR
 - ii) ARMA
 - iii) ARIMA
4. Briefly explain the different network methods used in water resources projects.
5. How the simplex method of linear programming is used in water resources? What are the advantages and limitations of the method?
6. Explain the following :
 - a) Flood frequency Analysis.
 - b) Risk analysis.
7. What is flood routing of a reservoir? Explain the basic flood routing equation and outline its method of solution.

OR

How will you calculate the optimum number of rain gauge stations in the basin for estimating mean rainfall?

8. Write short notes (Any Two) :
 - a) Recording type rain gauge
 - b) Updating of Network
 - c) Probable maximum Flood
 - d) Hydrology